1.Why would you advocate for bringing transparency or explainability in an ML/AI system? What are the advantages to the users as well as the organization providing that system?

Advocating for transparency or explainability is essential; because it helps people understand why a model made a particular decision, rather than treating it as a black box. Explainability builds trust, supports safety and enables responsible AI deployment.

Advantage to Users

1. Trust and Confidence: Users are more likely to rely on a system when they can understand its reasoning rather than blindly trusting predictions.
2. Ability to Contest or Correct Errors: If a system rejects a loan or flags a medical risk, users can challenge decisions with evidence.
3. Fair Treatment: Transparency helps detect discriminatory or unfair behavior , protecting users from bias.

Advantage to Organizations

1. Regulatory Compliance: Many industries (finance, healthcare, government) *require* explainability to meet legal standards.

2. Improved Model Quality: Explanations help engineers detect model weaknesses, biases, and data errors.

3. Accountability and Risk Reduction: Transparent systems lower the risk of harmful decisions, reputational damage, and lawsuits.

4. User Adoption: Products with clear reasoning gain higher user trust and wider acceptance.

Explainability is therefore both an ethical obligation and a practical business advantage.

2. What is a "fair" system? Take an example of an ML/AI system and tell us what achieving "fairness" would look like.

A fair system is one whose predictions or decisions do not systematically disadvantage individuals or groups based on sensitive attributes such as race, gender, disability, or age *unless legally and ethically justified*. A fair ML system ensures equitable outcomes or opportunities across groups.

Example

Credit approval prediction model.

What achieving fairness would look like:

- The model does **not** use protected attributes (e.g., race, gender) to make decisions unless justified.
- Approval rates for applicants with similar financial profiles are similar across demographic groups.
- Performance metrics (accuracy, false rejection rates) are balanced across groups - no one group faces disproportionate denials.
- Bias audits show no adverse impact, and explanations reveal that decisions rely on meaningful financial variables (income, loans), not proxies for sensitive attributes.

Fairness = equitable treatment + justified differences + continuous monitoring.

3. How do you see concepts like bias, fairness, accountability, and transparency-related?

Bias, fairness, accountability and transparency are interconnected pillars of responsible AI:

- Bias is the presence of systematic errors or unfair patterns learned from data.
- Fairness aims to reduce or eliminate those biases to ensure equitable outcomes.
- Transparency makes the system's reasoning visible, allowing stakeholders to detect where bias may be occurring.
- Accountability ensures there are people and processes responsible for monitoring, correcting, and justifying model behavior.

Together, transparency reveals bias, fairness corrects it, and accountability ensures ongoing oversight.

4. Give an example of a system that is biased but fair.

Example: Medical risk stratification system

A model may treat older patients as higher risk for certain procedures (e.g., heart surgery). This is technically biased toward age, but it is considered fair because age is a clinically valid predictor of medical risk, and the bias improves patient safety.

Other examples:

- College admissions giving preference to first-generation students (bias toward a group, but fairness for equity).
- Fraud detection systems that evaluate accounts differently based on transaction history.

In these cases, bias is purposeful and justified, making the system biased yet fair.